Activity 4.3 – CPI Counter

1.Design Approach:

Designed a processor performance monitoring module using Verilog HDL to calculate clock cycles, retired instructions and cycles per instruction (CPI). Implemented counters and arithmetic logic for CPI calculation.

2.Verification Steps:

- Created a separate testbench.
- Applied reset.
- Generated retirement signals.
- Checked clock cycle counting.
- Verified retired instruction counting.
- Tested CPI calculation.
- Simulated using ModelSim.

3.Results:

The testbench checks cycle counting, instruction retirement and CPI calculation. PASS/FAIL messages are displayed.

4.Issues Faced:

Counters must increment correctly. Division by zero must be avoided. CPI calculation requires proper arithmetic sizing.

5.Conclusion:

Successfully designed and verified a CPI monitoring module using Verilog HDL.